

Kai Love Cheung

Irvine, CA | kailovecheung@gmail.com | (425)305-8020 | [LinkedIn](#) | [Portfolio](#)

EDUCATION

University of California, Irvine

B.S. Electrical Engineering | Specialization in Digital Signal Processing

Spring 2027

GPA: 3.9

AICE Diploma | Received 2023

Cambridge Merit | Received 2021, 2022, 2023

PROJECTS

Custom Portable MP3 Player

July 2025 - Present

- Designed an MP3 player from scratch using a FireBeetle ESP-32 and selected components to provide full functionality including an audio jack, OLED screen, and capacitive touch buttons
- Optimized the power management system, integrating a rechargeable LiPo battery with onboard charging and engineered solutions to increase battery life
- Engineered DAC power delivery using a boost converter and MOSFET switching, addressing challenges with stable 5V regulation from a 3.7 V supply

[Splitty: Ergonomic Keyboard Design & Fabrication](#)

June 2025 - July 2025

- Designed and fabricated a fully wireless split ergonomic mechanical keyboard (“splitty”) by creating custom PCBs in KiCad, modeling the case and plate in Fusion 360
- Built and validated the keyboard matrix, scaling from single-switch schematics to a full 6x4 layout with thumb keys and ghosting prevention
- Implemented custom firmware using ZMK with device tree configurations, key matrix definitions, and multi-layer keymaps, enabling seamless Bluetooth communication
- Invented a columnar, tented, and unstaggered layout inspired by Lily58 but customized for comfort, aesthetics, and portability

Embedded Control System Robot Arm | Zotbotics@UCI FABWorks

January 2025 - June 2025

- Assembled and wired a robotic arm, configuring and tuning stepper motors with appropriate microstepping for smooth, precise motion control
- Developed Arduino firmware to execute a rock-paper-scissors sequence on button press, unifying stepper and servo control into a single program
- Engineered a stable power system using a 24 V supply with 12 V and 5 V buck converters to safely power microcontrollers and servos, ensuring reliable operation

iPod Refurbishment Project | IEEE@UCI Lab

Completed January 2025

- Upgraded iPod Video (5.5th Gen) with 2000mAh battery solution and flash storage
- Added Bluetooth Functionality using generic 3rd party Bluetooth module and cleverly rewired existing iPod buttons for a seamless listening experience
- Demonstrated proficiency in soldering and system optimization

Guitar Pedals

September 2024-June 2024

- Built DIY fuzz distortion and reverb pedals by ordering specialized PCBs, researching suitable electronic components and assembling by hand
- Reverse-engineered a simple overdrive circuit, refining electronic filter design skills

LEADERSHIP EXPERIENCE

WSSDA Conference Speaker

Hyatt Regency Ballroom, Bellevue, WA

Speaker

2021

- Spoke at the annual Washington State School Directors Association conference, participating in discourse between teachers and students
- Wrote a speech analyzing personal experience with research skills-based education and responded to live questions and criticism

PROFESSIONAL AFFILIATIONS

Institute of Electrical and Electronics Engineers @UC Irvine (IEEE@UCI)

2023-Present

ZiMS @UC Irvine (Zotbotics member)

2024-Present

PROFESSIONAL EXPERIENCE

Pedalheads

6046 West Lake Sammamish Pkwy NE, Redmond, WA

Bike Instructor

June 2024 – August 2024

- Coached children through progressive riding lessons, focusing on self-awareness, skill development, and safety abilities
- Developed leadership, collaboration, and problem-solving skills by managing a 2:10 instructor-to-student ratio
- Prepared for and addressed various challenges under pressure, collaborating with coworkers to ensure student safety

Cyan Swim

6601 132nd Ave NE, Kirkland, WA

Swim Instructor

January 2023 – August 2023

- Coached children individually, teaching foundational swimming skills up to competitive swim team strokes. Proudly taught several students from zero experience to swimming alone, without help
- Led and planned swim team practices, fostering an environment for improvement of all swimming skills for the team

SKILLS

Technical Skills: KiCad, Fusion, Microsoft Office, Datasheets, Verilog, Music Production, Passive & Active Filter Design, SPICE Simulation, Arduino, Soldering

Soft Skills: Problem-solving, Adaptability, Interpersonal Skills, Collaboration, Public Speaking, Leadership